

AZURE ENTERPRISE INFRASTRUCTURE LAB

06 — Identity & Security

Managed Identity + Azure RBAC + Key Vault

Evidence-based implementation documentation with attached Azure Portal screenshots

1. Objective

Demonstrate Azure identity and security fundamentals by reviewing RBAC scope, enabling a system-assigned managed identity on the Windows VM, creating an Azure Key Vault with Azure RBAC, assigning the VM the Key Vault Secrets User role, and troubleshooting a data-plane permission error while creating a test secret.

2. Environment

Resource / Setting	Observed value
Subscription	My Test Azure subscription 1
Resource Group	rg-azure-enterprise_MM
VM	vm-web-MM
VM OS	Windows Server 2025 Datacenter
VNet	vnet-azure-enterprise-MM
Subnet	WebSubnet_MM
VM Private IP	10.0.1.4
VM Public IP	20.244.97.131
Key Vault	kv-azure-enterprise-mm
Key Vault region	Central India
Key Vault pricing tier	Standard
Key Vault permission model	Azure role-based access control

3. Security Architecture

- User / administrator → Azure RBAC at subscription/resource scopes
- vm-web-MM → System-assigned Managed Identity → Microsoft Entra ID
- Managed Identity → Key Vault Secrets User → kv-azure-enterprise-mm
- Key Vault → Secrets → DemoSecret (planned test secret; creation currently blocked by the user's own data-plane permissions)

4. RBAC Scope Review

The subscription IAM screenshot shows the signed-in account has an active Owner role at subscription scope. The VM IAM screenshot shows that Owner is inherited from the subscription.

No additional Owner or Contributor assignment is required for the administrator. This is useful evidence for explaining RBAC scope inheritance and least-privilege design.

5. Managed Identity Configuration

A system-assigned managed identity was enabled on vm-web-MM. The identity is tied to the lifecycle of the VM and can be granted Azure RBAC permissions without embedding credentials in code or configuration.

Verified state: System assigned = On.

6. Key Vault Creation

- Key Vault name: kv-azure-enterprise-mm
- Resource group: rg-azure-enterprise_MM
- Region: Central India
- Pricing tier: Standard
- Soft delete: Enabled
- Retention: 90 days
- Purge protection: Disabled
- Permission model: Azure RBAC
- Networking: Public endpoint / all networks for the initial lab

7. Key Vault RBAC Assignment

The VM managed identity was assigned the built-in role:

Key Vault Secrets User

Scope: kv-azure-enterprise-mm. Member: vm-web-MM (Virtual machine managed identity).

This role is intended for reading secret values; it is not sufficient for the signed-in administrator to create or manage secrets.

8. Troubleshooting: Secret Creation Forbidden

A test secret named DemoSecret was prepared with a dummy value, but creation failed with a Forbidden/RBAC error.

Observed issue: The portal reported that the operation was not allowed by RBAC and advised allowing several minutes for recently changed role assignments to become effective. The caller was not authorized to perform the secret set operation.

This behavior is consistent with a separation between Azure resource-plane administration and Key Vault data-plane permissions. Having Owner at subscription/resource scope does not by itself grant the required Key Vault secret data action under the Azure RBAC permission model.

9. Correct Remediation

To allow the administrator to create/manage test secrets while preserving the VM's least-privilege read access:

1. Key Vault → Access control (IAM) → Add role assignment

2. Role: Key Vault Secrets Officer
3. Assign access to: User, group, or service principal
4. Member: the administrator's Azure account
5. Review + assign
6. Allow several minutes for RBAC propagation, then refresh/retry

Do not remove the VM's Key Vault Secrets User role. Keep it so the VM can later retrieve secrets without write permissions.

10. Planned Test Secret

- Name: DemoSecret
- Value: AzureLab-Test-2026
- Enabled: Yes

Use this dummy value only. Never use a production password, API token, connection string, or other real credential in a learning secret or public GitHub repository.

11. Validation Plan

7. Create DemoSecret successfully from the administrator account after the Secrets Officer role propagates.
8. Keep vm-web-MM as Key Vault Secrets User.
9. From vm-web-MM, authenticate with its managed identity and perform a read-only secret retrieval test.
10. Capture successful role-assignment, secret listing, and read-validation screenshots.
11. For a production-style design, replace the public Key Vault endpoint with private connectivity and restrict network access.

12. Interview Skills Demonstrated

- Azure RBAC and scope inheritance
- Least privilege
- System-assigned managed identity
- Microsoft Entra ID integration
- Azure Key Vault
- Key Vault RBAC
- Secrets management
- Data-plane vs control-plane permissions
- RBAC propagation troubleshooting
- Credential-free VM-to-service authentication

13. Interview Questions

Q1. Why use a managed identity?

It allows an Azure resource such as a VM to authenticate to supported Azure services without storing application credentials.

Q2. What is the difference between control-plane and data-plane permissions?

Control-plane permissions manage the Azure resource itself; data-plane permissions govern operations against the data or objects inside the resource, such as Key Vault secret operations.

Q3. Why did the VM receive Key Vault Secrets User?

The VM only needs to read secrets for the planned test; it does not need permission to create or delete secrets.

Q4. Why did secret creation fail even though the administrator is Owner?

The Key Vault uses Azure RBAC for its data plane, so the administrator needs an appropriate Key Vault data role for secret creation.

Q5. Why choose Key Vault Secrets Officer instead of Administrator?

Secrets Officer is narrower and supports secret management without granting all Key Vault data operations.

Q6. Why is public network access acceptable here?

It is being used for the initial learning exercise; a production design should consider network restriction/private connectivity.

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15. Screenshot Evidence

Figure 1 — Subscription IAM — Owner assignment

Microsoft Azure portal interface showing the 'My Test Azure subscription 1 | Access control (IAM)' page. The page displays the 'Check access' section, which includes a 'Check access' button and a message: 'Review the level of access a user, group, service principal, or managed identity has to this resource. Learn more.' Below this, the 'Mizanur Mollah Assignments' section shows a table with columns: Role Name, Scope, Membership, Condition, and Action. The table lists one assignment: 'Owner' with a scope of 'This resource' and membership of 'Mizanur Mollah'. The table also shows counts for Active (1), Eligible (0), and Deny (0) assignments. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Security, Resource visualizer, Events, Resource groups, Resources, Cost Management, Billing, Settings, Monitoring, and Help. The bottom status bar shows the Windows taskbar with the search bar, taskbar icons, and system tray information including temperature (30°C), time (14:03), and date (02-09-2026).

Role Name	Scope	Membership	Condition	Action
Owner	This resource	Mizanur Mollah	No	-

Subscription IAM — Owner assignment

Figure 2 — VM IAM — inherited Owner role

The screenshot shows the Microsoft Azure portal interface. The main content area displays the 'vm-web-MM' resource page with the 'Access control (IAM)' tab selected. The page shows the 'Owner' role assignment for the 'Subscription (inherited)' scope. The 'Role assignments' table lists the 'Owner' role with the description 'Grants full access to manage all resources in the subscription'. The 'Scope' is 'Subscription (inherited)'. The 'Group assignment' is 'Mizanur Mollah assignments - vm-web-MM'. The 'Condition' is 'None'.

Role	Description	Scope	Group assignment	Condition
Owner	Grants full access to manage all resources in the subscription	Subscription (inherited)	Mizanur Mollah assignments - vm-web-MM	None

VM IAM — inherited Owner role

Figure 3 — VM Network Settings — existing NSG/network evidence

The screenshot shows the Microsoft Azure portal interface for the VM 'vm-web-MM'. The left sidebar contains navigation options like 'Tags', 'Diagnose and solve problems', 'Monitor', 'Resource visualizer', 'Connect', 'Bastion', 'Windows Admin Center', and 'Networking'. The 'Networking' section is expanded, showing 'Network settings'.

The main content area displays the 'Network settings' for 'vm-web-MM'. It includes a dropdown for 'Network interface / IP configuration' set to 'vm-web-mm993 (primary) / ipconfig1 (primary)'. Below this, the 'Essentials' section lists various settings:

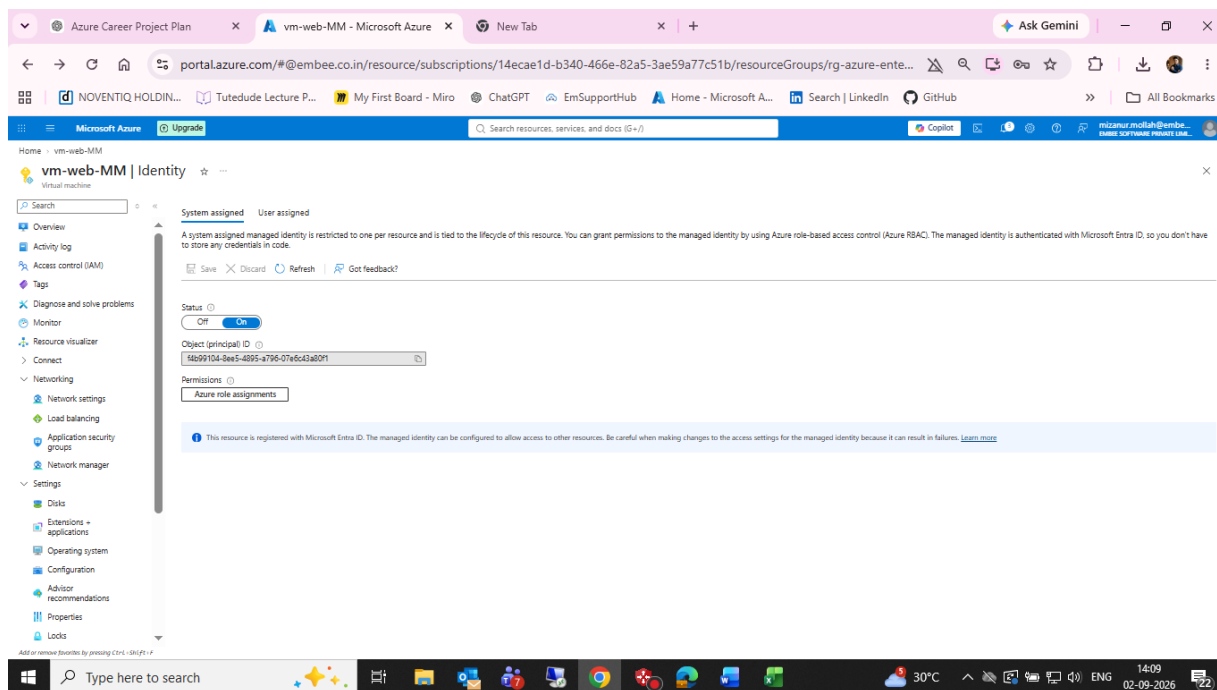
- Network interface: vm-web-mm993
- Virtual network / subnet: vnet-azure-enterprise-MM / WebSubnet_MM
- Public IP address: 20.244.97.131
- Private IP address: 10.0.1.4
- Admin security rules: 0 (Configure)
- Load balancers: 0 (Configure)
- Application security groups: 0 (Configure)
- Network security group: vm-web-MM-nsg
- Accelerated networking: Enabled
- Effective security rules: 0

The 'Rules' section is expanded, showing a table of network security group rules for the 'nsg-web-MM' group. The table has columns for Priority, Name, Source, Destination, Protocol, Port, and Action.

Priority	Name	Source	Destination	Protocol	Port	Action
100	Allow-HTTP	Any	Any	TCP	80	Allow
110	Allow-HTTP1	Any	Any	TCP	443	Allow
120	NEWACC	Any	Any	TCP	3389	Allow

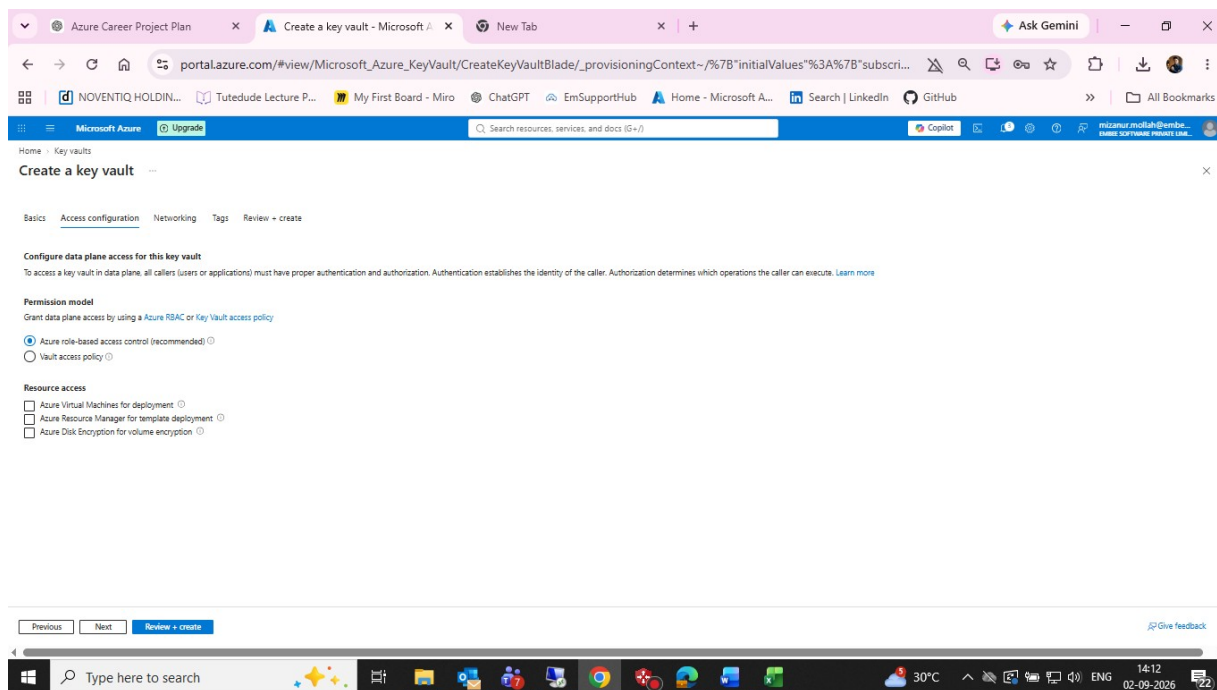
VM Network Settings — existing NSG/network evidence

Figure 4 — VM Identity — system-assigned managed identity enabled



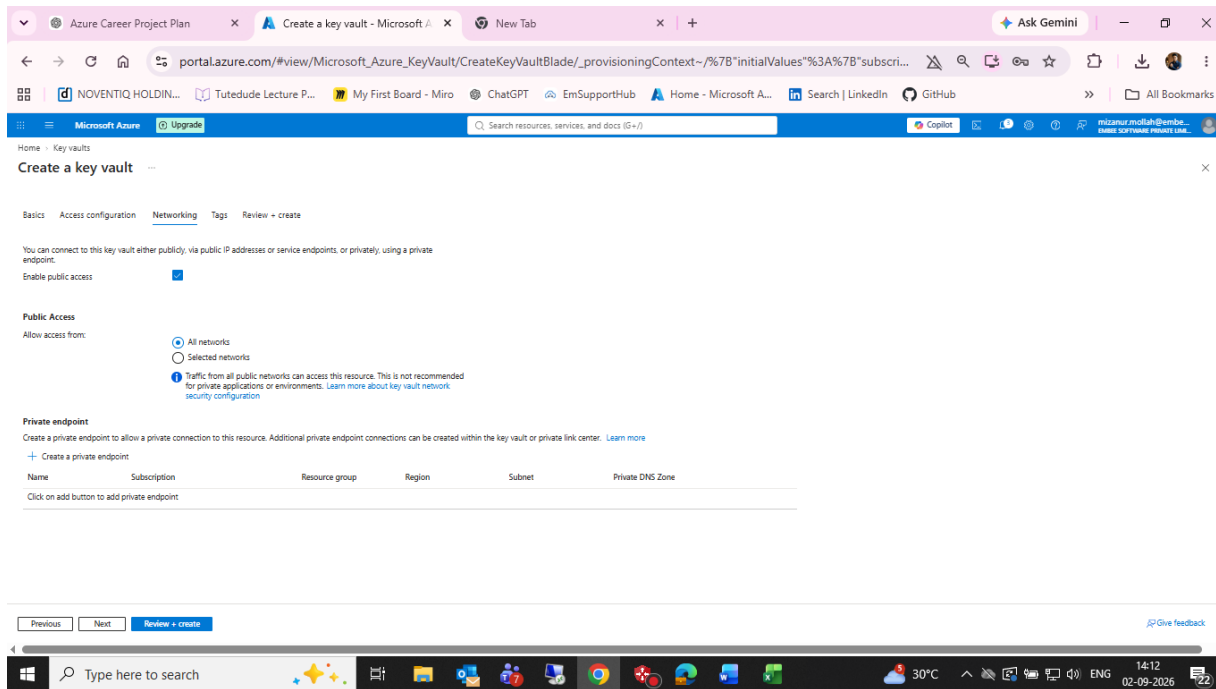
VM Identity — system-assigned managed identity enabled

Figure 5 — Key Vault Access Configuration — Azure RBAC



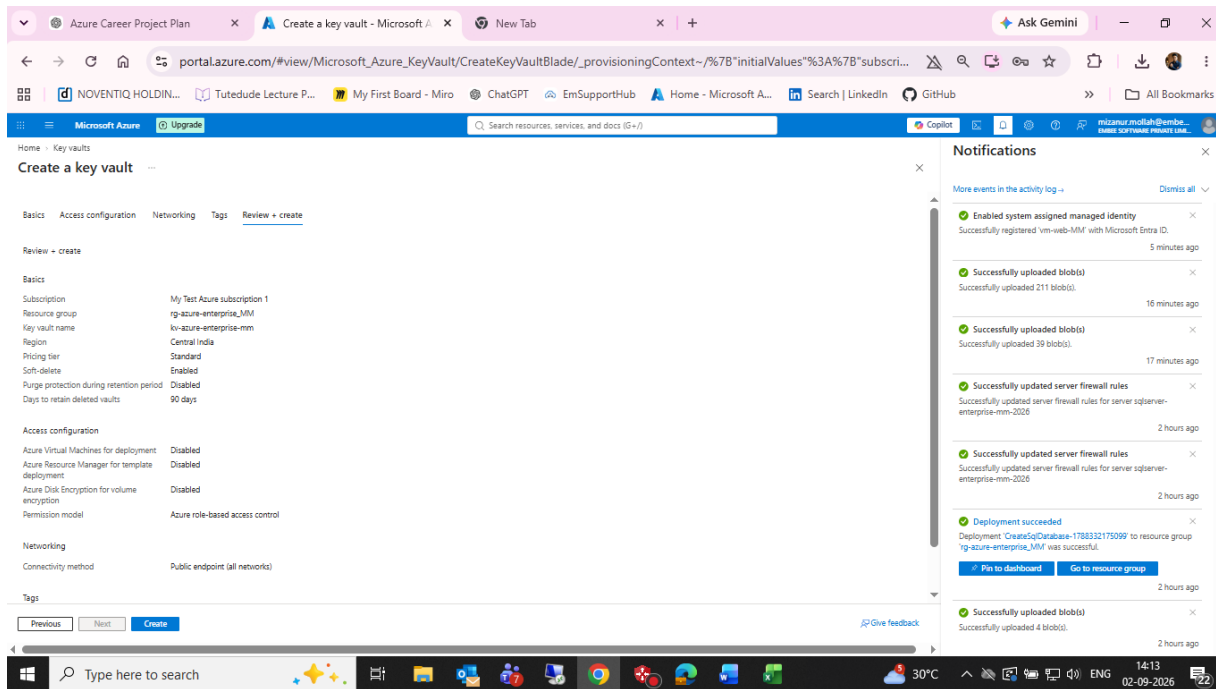
Key Vault Access Configuration — Azure RBAC

Figure 6 — Key Vault Networking — public endpoint, all networks



Key Vault Networking — public endpoint, all networks

Figure 7 — Key Vault Review + Create



Key Vault Review + Create

Figure 8 — Key Vault deployment operation details

The screenshot displays the Azure portal interface for a deployment operation. The main view is titled "kv-azure-enterprise-mm | Overview". The deployment is in progress, with a status of "Running". The deployment details table shows the resource "kv-azure-enterprise-mm" of type "Key vault" with a status of "OK". The operation details panel on the right provides a comprehensive overview of the deployment process, including the provisioning state, timestamp, duration, tracking ID, service request ID, status, type, resource ID, resource, extension, identifiers, and symbolic name. The notifications panel on the right lists various events, such as "Deployment in progress", "Enabled system assigned managed identity", "Successfully uploaded blob(s)", "Successfully updated server firewall rules", and "Deployment succeeded".

Deployment details

Resource	Type	Status
kv-azure-enterprise-mm	Key vault	OK

Operation details

Provisioning state: Running

Timestamp: 9/2/2025, 2:14:32 PM

Duration: 3 seconds

Tracking ID: 31d6fdd7-922f-45dc-92ee-2ad39fc14d5e

serviceRequestid: 9b828e19-184d-4dcb-b9f4-8773669fc906

Status: OK

Type: Microsoft.KeyVault/vaults

Resource ID: /subscriptions/14ecae1d-b340-406e-82a5-3ae59a77c51b/resourceGroups/rg-azure-.../providers/Microsoft.KeyVault/vaults/kv-azure-enterprise-mm

Resource: kv-azure-enterprise-mm

Extension:

Identifiers:

Symbolic Name:

Notifications

- Deployment in progress... Running X
Deployment to resource group 'rg-azure-enterprise-mm' is in progress.
a few seconds ago
- Enabled system assigned managed identity X
Successfully registered vm-web-MM with Microsoft Entra ID.
6 minutes ago
- Successfully uploaded blob(s) X
Successfully uploaded 211 blob(s).
17 minutes ago
- Successfully uploaded blob(s) X
Successfully uploaded 39 blob(s).
18 minutes ago
- Successfully updated server firewall rules X
Successfully updated server firewall rules for server sqgwen-enterprise-mm-2025.
2 hours ago
- Successfully updated server firewall rules X
Successfully updated server firewall rules for server sqgwen-enterprise-mm-2025.
2 hours ago
- Deployment succeeded X
Deployment 'CreateSqDatabase-1788332175099' to resource group 'rg-azure-enterprise-mm' was successful.
2 hours ago

Key Vault deployment operation details

Figure 9 — Add Role Assignment — Key Vault Secrets User selected

Add role assignment

Role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Search by role name, description, permission, or ID

Name	Description	Type	Category	Details
Reader	View all resources, but does not allow you to make any changes.	BuiltInRole	General	View
Advisor Reviews Contributor	View reviews for a workload and triage recommendations linked to them.	BuiltInRole	None	View
Anyscale Platform Administrator Role	Provides full access to Anyscale Cloud and resources as well as admin features such as usage and resource notifications.	BuiltInRole	None	View
Anyscale Platform Contributor Role	Grants read/write access to Anyscale Cloud and resources	BuiltInRole	None	View
Anyscale Platform Reader Role	Read access to Anyscale Cloud and resources	BuiltInRole	None	View
API Management Service Contributor	Can manage service and the APIs	BuiltInRole	Integration	View
API Management Service Operator Role	Can manage service but not the APIs	BuiltInRole	Integration	View
API Management Service Reader Role	Read-only access to service and APIs	BuiltInRole	Integration	View
API Management Service Workspace API Develo...	Has read access to tags and products and write access to allow: assigning APIs to products, assigning tags to products and A...	BuiltInRole	None	View
API Management Service Workspace API Product...	Has the same access as API Management Service Workspace API Developer as well as read access to users and write access t...	BuiltInRole	None	View
API Management Workspace API Developer	Has read access to entities in the workspace and read and write access to entities for editing APIs. This role should be assign...	BuiltInRole	None	View
API Management Workspace API Product Manag...	Has read access to entities in the workspace and read and write access to entities for publishing APIs. This role should be assi...	BuiltInRole	None	View
API Management Workspace Contributor	Can manage the workspace and view, but not modify its members. This role should be assigned on the workspace scope.	BuiltInRole	None	View

Notifications

- Deployment succeeded
Deployment 'kv-azure-enterprise-mm' to resource group 'vg-azure-enterprise-mm' was successful.
Go to resource Go to resource group
2 minutes ago
- Enabled system assigned managed identity
Successfully registered 'vm-web-MM' with Microsoft Entra ID.
8 minutes ago
- Successfully uploaded blob(s)
Successfully uploaded 211 blob(s).
19 minutes ago
- Successfully uploaded blob(s)
Successfully uploaded 39 blob(s).
20 minutes ago
- Successfully updated server firewall rules
Successfully updated server firewall rules for server sqlserven-enterprise-mm-2026
2 hours ago
- Successfully updated server firewall rules
Successfully updated server firewall rules for server sqlserven-enterprise-mm-2026
2 hours ago
- Deployment succeeded
Deployment 'CreateSqlDatabase-178832175099' to resource group 'vg-azure-enterprise-mm' was successful.

Add Role Assignment — Key Vault Secrets User selected

Figure 10 — Add Role Assignment — vm-web-MM managed identity selected

The screenshot shows the 'Add role assignment' page in the Azure portal. The 'Members' tab is active, displaying a table with the following data:

Name	Object ID	Type
vm-web-MM	f4b99104-8ee5-4895-a795-07a6c3a80f1	Virtual machine

Below the table, the 'Description' field is labeled 'Optional'. The 'Assign access to' section has 'Managed identity' selected. The 'Notifications' panel on the right lists several events, including 'Deployment succeeded' and 'Successfully updated server firewall rules'.

Add Role Assignment — vm-web-MM managed identity selected

Figure 11 — Key Vault RBAC role assignment success

The screenshot displays the Azure portal interface for the 'kv-azure-enterprise-mm' resource. The 'Access control (IAM)' section is active, showing a list of roles. The 'Role assignments' tab is selected, and a table lists various roles with columns for Name, Description, Type, Category, and Details. The 'AcrDelete' role is highlighted. On the right, a 'Notifications' panel shows several success messages, including 'Added Role assignment', 'Deployment succeeded', 'Enabled system assigned managed identity', and 'Successfully uploaded blob(s)'. The bottom of the screen shows the Windows taskbar with the search bar and system clock.

Name	Description	Type	Category	Details
Owner	Grants full access to manage all resources, including the ability to assign roles in Azure R...	BuiltInRole	General	View
Contributor	Grants full access to manage all resources, but does not allow you to assign roles in Azur...	BuiltInRole	General	View
Reader	View all resources, but does not allow you to make any changes.	BuiltInRole	General	View
Access Review Operator Service R...	Lets you grant Access Review System app permissions to discover and revoke access as n...	BuiltInRole	None	View
AcrDelete	acr delete	BuiltInRole	Containers	View
AcrImageSigner	Planned DEPRECATION on March 31, 2023. Grant the signing permission for content trus...	BuiltInRole	Containers	View
AcrPull	acr pull	BuiltInRole	Containers	View
AcrPush	acr push	BuiltInRole	Containers	View
AcrQuarantineReader	acr quarantine data reader	BuiltInRole	Containers	View
AcrQuarantineWriter	acr quarantine data writer	BuiltInRole	Containers	View
Advisor Discoverable Workload Re...	View discoverable workloads and their resources.	BuiltInRole	None	View
Advisor Discoverable Workloads C...	Convert discoverable workload into service group.	BuiltInRole	None	View
Advisor Recommendations Contri...	View assessment recommendations, accepted review recommendations, and manage the...	BuiltInRole	None	View
Advisor Recommendations Contri...	View assessment recommendations, accepted review recommendations, and manage the...	BuiltInRole	None	View
Advisor Reviews Contributor	View reviews for a workload and triage recommendations linked to them.	BuiltInRole	None	View

Key Vault RBAC role assignment success

Figure 12 — Create Secret — DemoSecret form

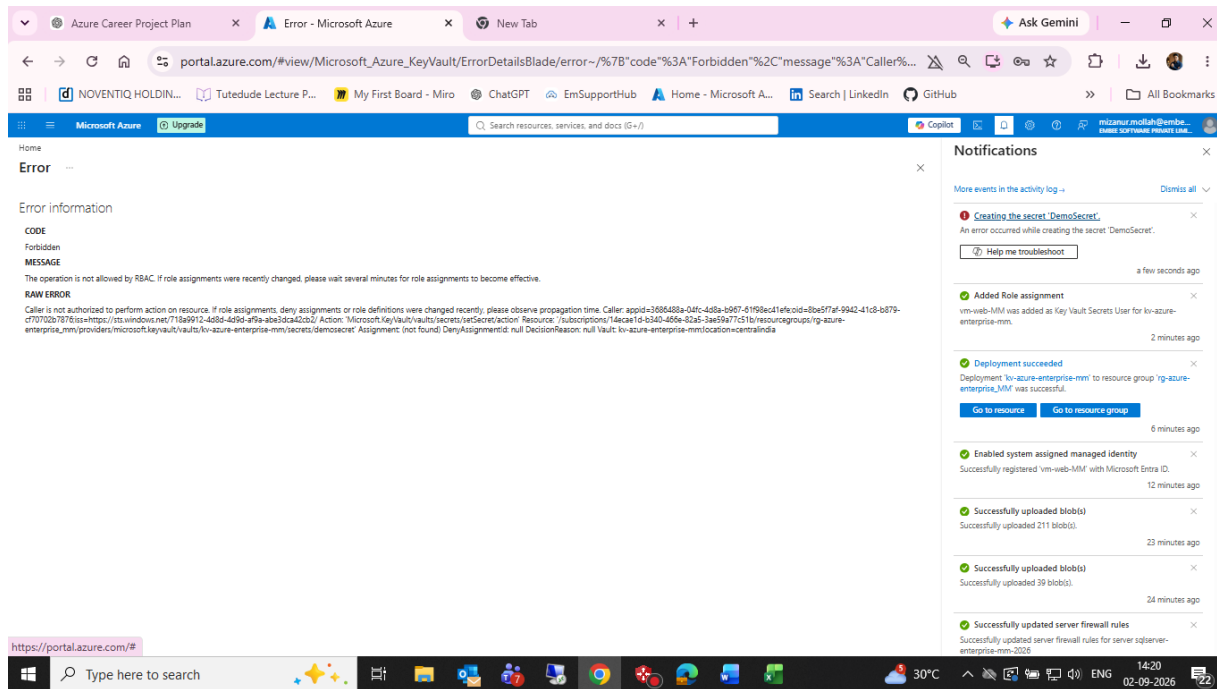
The screenshot shows the 'Create a secret' form in the Microsoft Azure portal. The form is titled 'Create a secret' and is located under the 'Secrets' section of the 'kv-azure-enterprise-mm' resource. The form includes the following fields and options:

- Upload options:** A dropdown menu set to 'Manual'.
- Name:** A text input field containing 'DemoSecret'.
- Secret value:** A text input field containing a masked value (represented by dots).
- Content type (optional):** A text input field.
- Set activation date:** A checkbox that is unchecked.
- Set expiration date:** A checkbox that is unchecked.
- Enabled:** A toggle switch set to 'Yes'.
- Tags:** A section with '0 tags'.

At the bottom of the form, there are 'Create' and 'Cancel' buttons. To the right of the form, there is a 'Notifications' panel showing a list of recent events, including 'Added Role assignment', 'Deployment succeeded', 'Enabled system assigned managed identity', and 'Successfully uploaded blob(s)'. The Windows taskbar at the bottom shows the time as 14:20 on 02-09-2026.

Create Secret — DemoSecret form

Figure 13 — Key Vault secret creation — Forbidden/RBAC error



Key Vault secret creation — Forbidden/RBAC error

16. GitHub Structure

- 06-identity-security/README.md
- 06-identity-security/screenshots/
- 06-identity-security/managed-identity/
- 06-identity-security/key-vault-rbac/

17. Git Commit

```
cd /d C:\azure-enterprise-infrastructure-lab
```

```
git add 06-identity-security
```

```
git commit -m "Document Azure identity, RBAC and Key Vault lab"
```

```
git push
```

18. Completion Checklist

- ☒ Subscription RBAC reviewed
- ☒ VM RBAC inheritance reviewed
- ☒ System-assigned managed identity enabled
- ☒ Key Vault created
- ☒ Azure RBAC permission model selected
- ☒ Key Vault Secrets User assigned to vm-web-MM

- ☒ RBAC assignment success captured
- ☒ DemoSecret creation attempted
- ☒ RBAC Forbidden error captured
- ☐ Assign Key Vault Secrets Officer to administrator
- ☐ Wait for RBAC propagation and retry DemoSecret creation
- ☐ Verify DemoSecret appears in Secrets
- ☐ Read DemoSecret from vm-web-MM using managed identity
- ☐ Capture post-success screenshots
- ☐ Update README with final validation
- ☐ Sanitize personal identifiers before public GitHub upload

19. Final Status

The core identity and authorization path is configured: vm-web-MM has a system-assigned managed identity, and that identity has Key Vault Secrets User access to kv-azure-enterprise-mm. The remaining blocker is the administrator's own permission to create secrets. The documented remediation is to assign Key Vault Secrets Officer to the administrator, allow RBAC propagation, create the dummy secret, and then validate read access from the VM.